# Tech Warmup II - Generative AI as an Engineering Tool

# Raw Research

Research Areas

- Slot machine logic(Rng, Rtp, volatility)
- Gameplay mechanics(wilds, multipliers, bonuses)
- Visual design & sound effects
- User types and behaviors

Game Features Identified

- Progression systems & rewards
- Bonus events and random triggers
- Leaderboards & social elements
- Custom themes and visual styles

User Research

- Casual players → simple & low-risk gameplay
- Social players → interaction & competition
- Relaxation users → stress-free experience

Persona Document:
Persona 1 - Casual Player
Timmy(20)

- Wants fun, quick gameplay
- Prefers simple controls
- Avoids complex interfaces

Persona 2 - Competitive Player

Chad(30)

- Wants to maximize winnings
- Interested in advanced features
- Plays longer sessions

User story:
As a casual player, I want simple controls so that I can start playing immediately.
As a competitive player, I want advanced features (multipliers, bonus rounds) so that the game stays engaging.
As a progression player, I want to unlock rewards so that I feel a sense of achievement over time.

## Personas

- ➔ Jonathan Shade 64 year old, retiree, overconfident after a divorce win. Wants big bets, pure luck, no progression systems (Owen Atis)
- ➔ Unnamed casual first-timer found the app through an ad, not a gambler, just wants light entertainment (Alex Perelson)

# Research Overview

## USER STORIES

- → As a chronic gambler, I want to place my own bets with massive returns so I win more than I started with
- → As an older player, I want fast, flashy, clearly communicated plays so I know exactly when I win or lose
- → As a bored young person used to modern games, I want overstimulation and instant gratification

## SLOT MACHINE

- → Domain/Functionality: The Slot Machine will include a progression based feature that can entice casual players to engage in the slot machine, paired with theming that is freely chosen by the user to give them the feeling of choice.
- → Visuals: The Slot Machine will incorporate themes that have distinct variations from one another to broaden appeal. Specific bright colors shall also be used to help in communicating intended reactions.
- → Audio: The Slot Machine will incorporate an ambiance track and specific sound effects that help in eliciting specific feelings.
- → Code Specifications: Code should be considered “clean” meaning that the architecture, comments, avoidance of duplicate functions and corresponding test cases must be implemented.

# AI Plan & AI Log

Tools and Approach:

- Using OpenAI Codex as our main tool
- Work in small steps, not one big prompt
- Keep same changes unless we clearly log changes

3-Step Process:

1. Build Base Version
   a. Generate simple working slot machine (HTML, CSS, JS)
2. Improve with Research
   a. Make it fun, casual, low-risk
   b. Add visuals, feedback, progression
3. Refactor & Test
   a. Clean code and better structure
   b. Handle edge cases
   c. Add comments and basic tests

How We Use AI:

- Prompt → Test → Log Results
- Track: what worked, failed, learned

Goal: Build a fun app while showing a clear, iterative AI workflow

Prompt Examples:

Start:

Create a slot machine app that uses vanilla web technology like HTML, CSS, JavaScript, and platform APIs. The slot machine should make fun of AI, as in you are winning tokens and spending tokens.

Run #3

Review this slot machine app and improve its software engineering quality. Keep working features intact unless necessary. Refactor for meaningful names, smaller functions, less duplicate code, and clearer modular structure. Add JSDoc comments with type annotations for the main JavaScript functions. Help handle simple edge cases and errors. Add or suggest basic unit tests for the core game logic and token system. Also help prepare the code so it can pass linting and validation checks for HTML, CSS, and JavaScript.

Run #10

Take this application and make the following changes: For the How to Play and Mechanics Explanation section, Give it its own Icon and make it minimizable, have it as a modal that is clickable by the user in order to prompt a pop-up instead of it constantly being visible. For the theme switch section, give it its own area on the left side. For the risk board, add the icons associated with the conditions. Include a settings bar that allows the user to change the theme and volume of the game. Remove the text underneath each symbol of the reel. Universal Icons for specific case point combinations, as an example the Cherry, Diamond and Lucky 7 symbols are shared across all themes. Change the color of “Theme Slot Cabinet” so that it is more visible. Change the music based off of the given theme that is currently selected. Ensure that the theme is not 12 notes repeated constantly.

# The 20 Runs - Key Miliestions

## Run 1-2

Token Furnace 3000. Basic slot machine with AI token theme, colorful UI, intro copy.

## Run 3

game-logic.js extracted. JSDoc type annotations, modular pure functions, unit tests added.

## Run 4-5

Theme system added (Lantern Festival, Riviera Postcards, Oasis Caravan). Per-theme symbols and jackpot rules.

## Run 6

Vault mechanic introduced. 24-spin bonus threshold, 150-token reward. Progress can drop backwards or fully reset on bad combos.

## Run 7-9

Mechanics panel, side layout. Web Audio API sound engine per-theme ambient music and per-action sound effects.

## Run 10-12

Help modal with live rules added, renamed to Lucky Spin Casino, and game-modes.js introduced 3-Reel & 5-Reel modes

## Run 13-16

Spin animation with reel-window CSS, mode locking during spin, performance-safe getNow() fallback, serif/display fonts introduced.

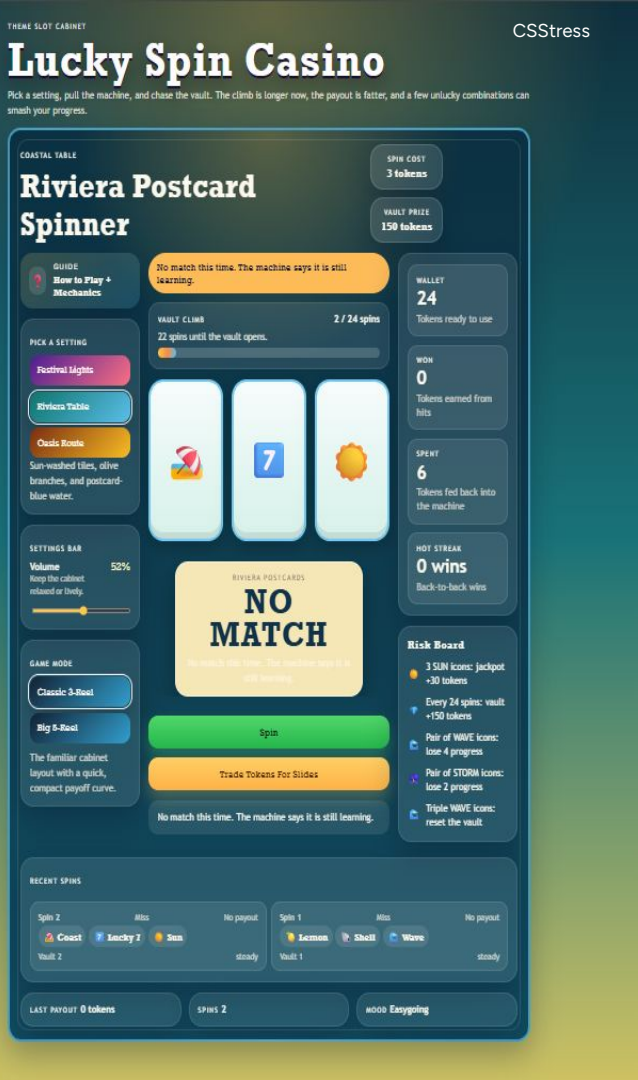
## Run 17-20

Recent spins history panel added. Symbol visual tuning. ARIA improvements. CSS grid layout fixes for reel centering.

# Lucky Spin Casino (Final Product)

- **2 Game Modes:** Classic 3-Reel (jackpot needs 3 matches) and Big 5-Reel (jackpot needs 5 matches)
- **3 Themes**:
  - Lantern Festival
  - Riviera Postcards
  - Oasis Caravan

  unique symbols and ambient audio per theme.
- **Vault Mechanic:** 24-spin climb to 150-token prize  progress can slip backwar or fully reset on bad combos
- **Audio**: Web Audio API  theme-based ambient music and per-action sound effects
- **Code Quality:**
  - game-logic.js (503 lines)
  - game-modes.js (42 lines)
  - script.js (1903 lines)
  - 3 test files (254 lines total)



# Takeaways

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|---|---|---|
| 01 | Precision beats luck | Specific prompts with exact line numbers and device constraints gave clean targeted changes |
| 02 | AI excels at features, struggles at layout | Audio and themes built fast. Reel centering took 7+ attempts across runs 7–20 |
| 03 | Testing is non-negotiable | Modular architecture from run 3 kept the engine stable through 17 more iterations |
| 04 | Context compounds | Feeding the previous run's changes into the next prompt prevented Codex from regressing |
| 05 | Human judgment is the real driver | Every improvement came from a team member observing the product and asking the right question |